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Beyond research delays: science funding disruptions limit community participation in climate and infrastructure planning



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The current U.S. federal administration has sought to intervene into every aspect of academic life, university functioning, and the research enterprise including by attacking academic freedom and integrity and canceling and retreating from publicly funded research. Such actions have profound adverse effects on the U.S. public, especially its most marginalized communities, and on science, itself. This perspective provides a telling example of such impacts through our own experience of funding cancellation, the disruptions it causes and the effects it has on urban systems and the communities they support. By focusing on our project that sought to center environmental justice communities in urban transportation and climate planning we offer insights into the wide-ranging effects of such disinvestment, including on sustainability and air quality efforts, with recommendations for moving forward to advance sustainable, equitable, and resilient cities.

Poor air quality contributes to billions of dollars in lost economic activity, adverse health outcomes, including asthma¹, cardiometabolic risk², cardiovascular disease³, and poor mental health⁴, and an increasing number of premature deaths annually^{5–7}. Residents of urban areas suffer some of the worst air quality on the planet due, in large part, to emissions from transportation – the movement of people, goods, and services across the surfaces of the most populated places^{8,9}. Transportation continues to be the leading cause of poor air quality in urban areas (e.g., refs. 10,11). The American Lung Association estimates that over half of U.S. residents live in areas with unhealthy air¹², with the majority residing in urban areas¹³. Many of these people live and work in California, which has 7 of the 10 most air-polluted places, and 13 out of 25 such places, in the country¹². According to the US EPA, Sacramento, Los Angeles, and San Francisco alone have experienced some of the most polluted air in the past decade¹³.

In addition to poor air quality, transportation emissions continue to drive climate change, contributing an estimated 9 billion tons of carbon into the atmosphere each year globally¹⁴, making it an essential area of study for urban sustainability. As cities continue to grow in both population and area size, they are set to increasingly contribute to both poor air quality and climate change¹⁵. Addressing poor air quality can help advance the U.N. Sustainable Development Goals (SDGs) – including Goal 3 by promoting good health and well-being and Goal 11 by developing sustainable cities and communities¹⁶. Enhancing our ability to identify and quantify transportation emissions can inform better decision-making on reducing contributors to both poor air quality and drivers of greenhouse gas emissions. Doing so can meaningfully promote good health and sustainable cities. To tackle both poor air quality

with its attendant consequences and to work towards sustainability, more research is needed to better understand the impacts of transportation on communities, particularly vulnerable ones, who tend to bear the brunt of such environmental inequality¹⁷, resulting in widening disparities¹⁸.

Situating our work

To gain a deeper understanding of these dynamics, we developed an interdisciplinary community-engaged project to examine the intersecting impacts of poor air quality, climate risks, and transportation infrastructure on vulnerable, urban communities in California. We proposed and were awarded U.S. federal funding to explore climate and transportation infrastructure issues from the perspective of community-based organizations and researchers through an assessment of risk perceptions, priorities, and data tools.

We aimed to utilize our expertise as qualitative and quantitative researchers, with extensive experience examining transportation planning, community engagement, and environmental justice issues^{19–30}, to conduct a multi-phased mixed-method study based in California (USA). Through the qualitative work, we intended to conduct semi-structured interviews with community-based organizations focusing on environmental justice to understand the needs and efficacy of current climate response strategies in partnership with local planners, policymakers, and engineers across the state. We then planned to build upon our findings on qualitative risk perceptions to develop a comparative quantitative and spatial analysis of vulnerability and risk. Through this quantitative component, we aimed to better understand how spatial dashboards for climate vulnerability can both address and fail to address residents' risk concerns.

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Importance of our approach

Partnering collaboratively with local community-based organizations focused on environmental justice, this project aimed to enhance methods for collecting community-level data and providing equitable opportunities for vulnerable, urban communities to provide input on climate infrastructure, including transportation and its attendant effects on air quality, and its harmful impacts. Furthermore, our goal with this mixed-methods work was to assess how the perceptions of risks in vulnerable communities align with existing secondary data tools for analyzing risks and priorities, including those related to air quality. Through this paired examination, we aimed to assess whether gaps exist among research, practice, and community perspectives on climate and environmental risk and benefit indicators, so that we can work towards improving risk assessment approaches by promoting policy-actionable metrics for decision-making aimed at reducing environmental injustices.

Despite the proposed and recognizable benefits of our work, progress on the project was disrupted when our U.S. federal funding was paused for four months into the start of our research activities and only released after legal intervention by universities and related academic organizations³¹. During the funding suspension, data collection was halted, research staff had to be reassigned or furloughed, and collaborations with local community-based organizations were interrupted. While funding has been reinstated as a result of a judicial ruling (about six months later), our team continues to face challenges as we work to restart operations, re-assign staff, and regain lost momentum. This has led to a situation where we are working within a severely condensed timeline, with limited ability to fulfill the project goals in the ways originally intended. However, like many other research teams in similar situations, our team is working to refine our methodology and remains committed to delivering impactful findings that can benefit our local community partners, as well as researchers and practitioners working on climate, infrastructure, and air quality topics more broadly.

More critically, these funding disruptions have not only caused issues for our research team and staff but will have profound impacts on our local community partners and community members in vulnerable urban areas across California. With our project's focus on incorporating community feedback on climate and infrastructure risks, we aimed to provide an opportunity for vulnerable communities to have their voices incorporated into planning research and the development of practical data tools. When environmental justice projects such as these are delayed or canceled, it provides one less avenue for understanding and working with communities to address these knowledge gaps to inform action to reduce such inequality in the first place.

These considerations are further exacerbated when considering the gaps in specific climate and infrastructure spaces, such as surrounding transportation and air quality topics and issues. Transportation is a field dominated by techno-rational planning, where quantitative metrics are privileged in the forecasting, design, and evaluation of facilities and services. Levels of service, vehicle miles traveled, frequency and efficiency, value of travel time, and crash outcomes are examples of common measures used to evaluate the performance of a transportation system. A similar focus on quantifying certain kinds of air pollutants in limited areas for regulatory and remediation purposes limits the incorporation of residents' lived experiences in regulatory decision-making and governance. These measures, from a utilitarian perspective, can help identify whether transportation investments and air quality impacts are economically efficient, effective in providing value to system users, reduce certain types of air pollutants, and provide an efficient use of resources for government investment. Cost-benefit analyses, for example, are common methods of accounting for the value of transportation investments.

Yet such a techno-rational, quantitative approach reduces inputs and effects of these investments to a dollar amount, which rarely captures the full distribution of benefits and burdens^{32,33}. Doing so misses both the experiences of those users and the communities through which such a system operates and the impacts this system has. But listening to and incorporating community knowledge into transportation systems can ameliorate such

blind spots. Community knowledge offers a distinct perspective on how to measure inputs and the impact of transportation on various environmental justice issues, including air quality. First, community perspectives matter. For example, a technical analysis might reveal that a walking route to a transit stop has no pedestrian crashes and low traffic volume, demonstrating that it is a safe route to walk. However, community knowledge might reveal that the street lacks lighting to illuminate the path at night and has limited visibility into surrounding homes and businesses, creating a place where people *feel* unsafe walking, regardless of what official statistics indicate. Similarly, regulatory air monitoring may look at certain hazardous air pollutants but only in one area with a limited scope of what and how much is detectable, whereas community members may be able to share both visual experiences of air pollutants as well as health experiences that may be tied to poor air quality (e.g., asthma, trouble breathing). Regulatory air monitoring is rarely performed at a scale that can account for these hazards from an environmental justice perspective. For example, researchers have investigated proximity to freeways and their effect on marginalized populations, but are only able to do so using proximity or modeled values at a county-level granularity^{34,35}. Even with these methodological limitations, they find that marginalized populations are increasing near major roadways with public health and environmental justice implications for these populations (e.g., ref. 34). Climate change is likely to exacerbate such inequalities, with frontline communities like these also the most socially vulnerable to climate impacts, including worsening air quality from a compounding array of sources (e.g., transportation emissions, wildfire smoke) (e.g., ref. 23).

Such methodological limitations, coupled with increased environmental justice impacts, necessitate the need for new approaches. Crucially, centering community perspectives is one approach that can improve methodologies and address environmental injustice, including poor air quality. Such an approach is a form of epistemic justice, where decision makers value the knowledge that communities possess as legitimate forms of data, on par with quantitative sources of data. Second, community knowledge illuminates intersectional forms of vulnerability that are challenging to measure or identify with solely quantitative data. Certain climate and environmental tools, such as the nationwide US Climate Vulnerability Index³⁶ and California's CalEnviroScreen³⁷, rely on perspectives from community-based organizations to identify relevant indicators that reflect environmental and climate injustices, the latter of which is a critical data tool used in state policy and funding decisions. These community-informed tools are perhaps more inclusive than other data dashboards that do not rely on community input but could also mask local conditions that are difficult or impossible to implement at neighborhood scales. This was precisely the gap in knowledge our federally funded work sought to fill.

Just as transportation is a means to access critical destinations and live a meaningful life, analyzing transportation and climate justice requires an approach that goes beyond the constraints of road infrastructure, transit services, and specific transportation policies. A transdisciplinary approach that includes urban studies and policy and environmental justice, along with engineering, public health, and planning, is needed to bring justice concerns and approaches into the transportation field to understand and address climate vulnerability and environmental justice. Doing so requires teams of researchers from across disciplines and fields to come together to create holistic and thoughtful approaches to such pressing questions.

Implications

The ongoing disruptions and cancellations to federally funded projects, such as ours, have short-, medium-, and long-term impacts across space and time. The short-term implications include a range of negative outcomes, such as layoffs, inaccessible data, insufficient data collection, and exclusion of community members from data decision-making. Medium-term implications range from the loss of graduate students - who are the next generation of scientists, analysts, and regulators - legislative staff, regulatory decisionmakers, and a trained civic workforce. Long-term implications include worsening impacts of climate change and increased environmental injustice - including worsening air quality - along with heightened epistemic

injustice, and all the adverse health, social, environmental, and economic outcomes that accompany these changes. In addition, there is a potential lag time for such negative impacts to appear in residents' daily lives, further delaying our progress toward climate goals, including the SDGs, while increasing pollution and producing adverse health and environmental outcomes.

There are also implications for knowledge gaps, including to specific fields such as air quality research, transportation and urban planning, sustainability and climate science, and community knowledge and awareness. Another concern is the quality of work that can and cannot get done with the federal disinvestment in and retreat from science, broadly, and infrastructure and its attendant effects, more specifically.

Questions like the ones investigated here will go unanswered as long as the continued assault on universities, academic integrity and freedom, and funding streams are maintained, proliferating a sense of uncertainty and anxiety. Their answers could lead to identifying what kinds of data are needed to assess climate injustice in transportation planning from frontline community perspectives, how to assess and measure climate risk and vulnerability, and what kinds of effects will decarbonization have on diverse urban populations, among many others. With the recent volatility in federal administration priorities over the past decade, the policy whiplash has created both opportunities and challenges. Such research in California is arguably more important than ever, given the recent pullback in science funding and pressing climate priorities under the current U.S. federal administration as of writing (2026). Such back-and-forth shifts in international policies and priorities on climate have also left global partners needing to recalibrate, with many no longer trusting U.S.-led climate interventions. Doing so has once again thrust California into a climate leadership role in the U.S. and globally on innovating and implementing progressive climate policies to advance sustainability and, crucially, studying the impacts of these policies on communities and environmental justice issues.

Recommendations and conclusions

Several conclusions and three major recommendations have emerged from this research context. First, defunding the Department of Transportation and its research arms has critical implications for communities across the U.S. and the range of issues they face, including poor air quality from transportation emissions. Second, it is clear that the integration of climate science and transportation is critical to improving air quality. This is particularly true now, given the collection of federal policies that increase greenhouse gas emissions and defund climate resilience, mitigation, and adaptation efforts. Examples include recent federal policy changes that attempt to relax emissions standards, rescind the rule (i.e., "endangerment finding") to account for greenhouse gas emissions in environmental decisions (e.g., ref. 38), removal of EV rebates, along with disbanding knowledge infrastructure through the removal of other offices related to climate science such as the White House Office of Science and Technology Policy. Finally, integration of transportation and environmental justice at the federal level was a watershed moment for acknowledging equity in the U.S. federal government. At the same time, tremendous work is being done across U.S. cities and states to advance climate equity while building resilient, sustainable, and equitable infrastructure. There is work to be done in collaborating with these state and city partners, as well as outside of formal state partnerships^{39,40}.

In conclusion, U.S. federally funded research is imperative for addressing poor air quality from transportation emissions. Such research can help put communities at the center of data gathering and reporting on environmental and climate indicators. Doing so can advance sustainable development goals, including SDG 3 to ensure healthy lives and SDG 11, making cities inclusive and sustainable. This work can help advance environmental justice across cities.

Data availability

There are no data associated with this perspective.

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Author contributions

S.C., C.E.B.C., and J.M.B. conceptualized, wrote, and revised the manuscript.

Competing interests

The authors declare no competing interests.

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